

Laxmi Charitable Trust's

Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce

Department of Information Technology (B.Sc.I.T Semester V)

MERN STACK

Practical –I

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Class : TYIT	Batch : 1
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1.Environment Setup and JavaScript Basics.

a. Install Node.js and Visual Studio Code, and verify installation.

```
PS C:\Users\IT-37\Desktop\MERN STACK> node -v
v24.17.0
PS C:\Users\IT-37\Desktop\MERN STACK> npm -v
Unknown command: "-"

To see a list of supported npm commands, run:
npm help
```

b. Create a simple Node.js file and run it using the terminal.

Code : console.log("hello")

console.log("Raj")

Output :

```
PS C:\Users\IT-37\Desktop\MERN STACK> node prac.js
hello
Raj
PS C:\Users\IT-37\Desktop\MERN STACK>
```

c. Write a Node.js script demonstrating basic JavaScript commands (console log, variables, and arithmetic operations).

Code :

let num = 20;

let num2 = 10;

console.log(num+num2);

console.log(num-num2);

console.log(num*num2);

console.log(num/num2);

console.log(num%num2);

Output :

```
PS C:\Users\IT-37\Desktop\MERN STACK> node prac1c.js
30
10
200
2
0
Rajj
```

d. Create and run a simple program using variables and functions.

Code :

```
// Variables
let num1 = 20;
let num2 = 20;
// Function
function add(number1, number2) {
    return number1 + number2 ;
}
// Function call
console.log("ADD",add(num1,num2));
```

Output:

```
PS C:\Users\IT-37\Desktop\MERN STACK> node prac1d.js
ADD 40
Rajj
PS C:\Users\IT-37\Desktop\MERN STACK> |
```

e. Demonstrate use of operators and control statements in JavaScript

Code :

```
// Variables
let num1 = 15;
let num2 = 10;

// Arithmetic Operators
let sum = num1 + num2;
let difference = num1 - num2;
let product = num1 * num2;
let quotient = num1 / num2;

console.log("Sum:", sum);
console.log("Difference:", difference);
console.log("Product:", product);
console.log("Quotient:", quotient);

// Comparison and Control Statement (if-else)
if (num1 > num2) {
    console.log(num1 + " is greater than " + num2);
}
```

```

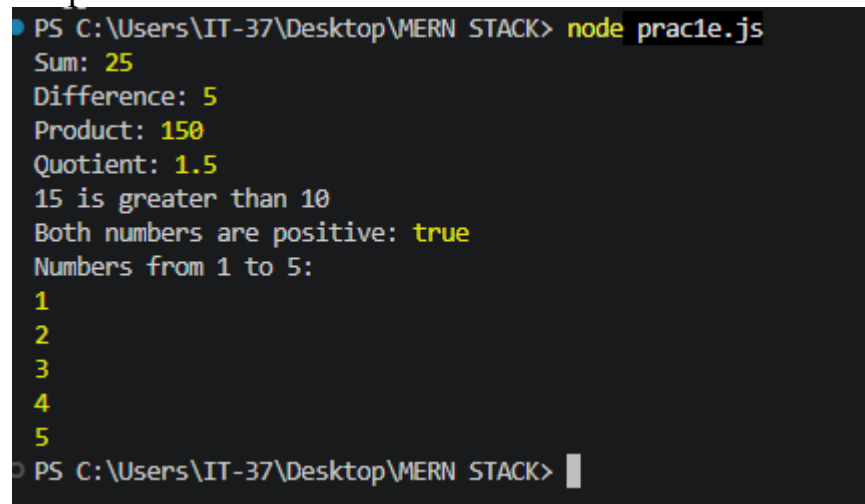
} else if (num1 < num2) {
    console.log(num1 + " is less than " + num2);
} else {
    console.log("Both numbers are equal");
}

// Logical Operator
let isPositive = num1 > 0 && num2 > 0;
console.log("Both numbers are positive:", isPositive);

// Loop Control Statement (for loop)
console.log("Numbers from 1 to 5:");
for (let i = 1; i <= 5; i++) {
    console.log(i);
}

```

Output:



```

PS C:\Users\IT-37\Desktop\MERN STACK> node prac1e.js
Sum: 25
Difference: 5
Product: 150
Quotient: 1.5
15 is greater than 10
Both numbers are positive: true
Numbers from 1 to 5:
1
2
3
4
5
PS C:\Users\IT-37\Desktop\MERN STACK>

```

f. Write programs using loops and template literals.

Code :

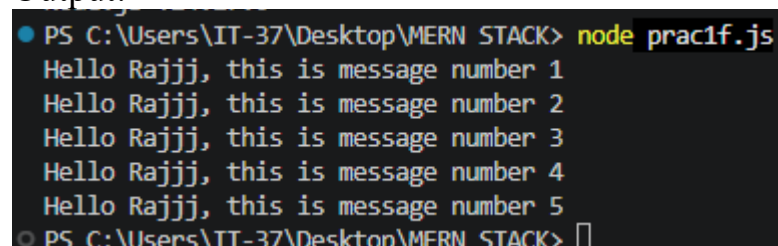
```

let name = "Rajjj";

for (let i = 1; i <= 5; i++) {
    console.log(`Hello ${name}, this is message number ${i}`);
}

```

Output:



```

PS C:\Users\IT-37\Desktop\MERN STACK> node prac1f.js
Hello Rajjj, this is message number 1
Hello Rajjj, this is message number 2
Hello Rajjj, this is message number 3
Hello Rajjj, this is message number 4
Hello Rajjj, this is message number 5
PS C:\Users\IT-37\Desktop\MERN STACK>

```